1 3. Data and Measurement

Our sample spans every FOMC meeting with a priced statement from January 2006 through April 2026 — 161 meetings, covering the pre-crisis tightening, the crisis and zero-lower-bound era, the forward-guidance window, the 2020 pandemic response, the 2022–2023 tightening cycle, and the current balance-sheet run-off. Three measurement blocks feed the paper: the shock decomposition, the statement text measures, and the fund-flow panel.

1.1 3.1 The shock decomposition

Policy surprises come from the Acosta USMPD family of high-frequency surprise measures (target and path shocks around FOMC announcements), extended through 2026 with USMPD-based construction after July 2022 (`combined_shocks_acosta_usmpd.csv`; 250 meeting events from 1995, of which 161 fall in our 2006+ sample window). We apply the market-inversion split of Section 2.2: a target shock is classified as the conventional component (MP) when its sign disagrees with the same-day CRSP value-weighted market return, and as the information component (CBI) when the signs agree. This one-line rule is the sign-restriction identification in its cleanest form — no auxiliary instruments, no high-frequency futures pricing beyond the construction of the underlying surprise — and it inherits the JK logic while remaining auditable from two columns of data.

Regime indicators mark (i) the forward-guidance era (December 2008 – December 2015; 56 of the 161 meetings), (ii) the 2020–2022 zero-rate window, and (iii) the post-2022 tightening cycle.

1.2 3.2 Statement text measures

Statements are scraped from the Federal Reserve's press-release pages (2006–2026) and cleaned to the canonical statement body (`data/fomc_statements_clean.json`; median length a few hundred words; the 2004–2005 short-statement era predates our window). Two text measures are used:

- **Dictionary tone.** LM hawkish-minus-dovish word counts per 100 words (`tone100 = sentiment × 100`). Naming discipline: this variable is never called "sentiment" (the codebook reserves that word out of the paper entirely to avoid collision with the investor-sentiment literature).
- **LLM score.** A rebuilt measure that reads the full statement text and returns a −100…+100 stance score with a one-line justification (161/161 statements scored; Section 7 reports its construction and the five-layer audit).

[SYSTEM_NOTE: Content compressed. Read the full version if needed.]on share changes; the flow window must start at the meeting month or later (`gap  3` calendar months); point-in-time Morningstar classification at the aggregation date; and flows bounded at 50% of TNA to contain outliers. The panel aggregates to 1,099 fund-month observations across seven asset classes (government bonds, corporate bonds, real assets, large-cap equity, developed-market equity, emerging-market equity, small-cap equity) over 157 meeting months.

1.3 3.4 One name, one definition

The variable codebook (`paper/codebook.docx`) enforces single definitions: shocks are innovations; "surprise" appears only in the GSS-construction context; tone is never sentiment; "information shock" only in JK-discussion context, quoted. Price reactions use one definition (CRSP value-weighted same-day return) unless a table header states otherwise. The audit trail for every headline number runs from the codebook through the archival result files (`price_results_v1.json`, `flow_results_rerun_v8.json`, `models_v2_results.json`) — numbers that could not be traced to an archival source were retired during drafting. # 7. Measurement: When the Dictionary Fails

Sections 4–6 used dictionary tone as the text measure. This section asks the question the tone literature usually skips: *how good is the dictionary at its job?* We answer by rebuilding a statement-based LLM score under a five-layer hallucination-audit protocol, and by showing that the dictionary's failures have a characteristic signature — resolution loss exactly when the Fed's communication paradigm shifts.

1.4 7.1 The rebuilt LLM measure

An earlier draft used an LLM score proxied from meeting metadata; it is retired. The rebuilt measure reads the full statement text and returns a −100…+100 stance score with a one-line justification (161/161 statements). Properties:

- Correlation with dictionary tone: **r = +0.351** — the two measures overlap but are far from redundant.
- Correlation with the CBI shock: **r = +0.027** — as Pillar 1 requires. The inversion outcome must not be a property of the text; if it were, the decomposition of Section 2.2 would be circular.
- No flow contribution (p = 0.17 in the Section 5 specification), mirroring tone — the null extends to every text measure.

1.5 7.2 Divergence is resolution loss, not bias

Eleven statements show |z(LLM) − z(tone)| > 2. On those statements the dictionary output is nearly constant (std = 1.10, mean +2.77) while the LLM separates directions sharply (std ≈ 71): Operation[SYSTEM_NOTE: Content compressed. Read the full version if needed.]lve the direction from the words at all.

1.6 7.3 The five-layer audit

LLM measures invite a legitimate objection — hallucination. We therefore ported a five-layer defense (retrieval consistency, vector fact-check, CROWN anchor-independence, multi-node voting, self-consistency) from our earlier agent infrastructure to the scoring task, and audited a 30-statement sample (11 divergences + 19 random, seed 42):

- Retrieval consistency: 53% word-level, 73% embedding-match (own vs random statement); weak margins reflect the template nature of FOMC prose, not fabrication.
- CROWN: showing the auditor the dictionary score does not anchor the LLM (0/30 triggered) — the measures are independent, as claimed.

- Multi-node vote, prompt variants: 100% direction consensus after fixing a float-threshold bug (2/3 = 0.6667 vs a 0.667 threshold had rejected all two-of-three consents — 9 false "LikelyFalse" verdicts that were an audit bug, not an LLM failure).
- Self-consistency (T = 0.3, 3 samples): 70% pass; 2 borderline statements flagged.
- **Cross-vendor layer**: GPT-4.1-mini, Claude-haiku-4.5, DeepSeek-flash, Gemini-2.5-flash (thinking models; reasoning tokens count against `max_tokens` — operational finding). Pairwise r = 0.89–0.97; sign-level agreement 93.1% (27/29); **on the 11 LM-divergence statements, sign agreement across vendor families is 100%**.

The audit caught exactly one real error: on the Operation Twist statement (2011-09-21) the main router returned +85 (hawkish) once; three reruns returned −70 each time, and all four vendor families independently returned strongly dovish scores (−60 to −90). "Extend the average maturity of its holdings" is an easing operation; the correction stands, the score file carries the audit note, and the Section 4 correlations moved r = 0.335 → 0.351.

1.7 7.4 Positioning

The LLM measure is a **diagnostic of dictionary failure, not a better scorer**. It adds no price or flow predictiveness (Sections 4–5), so we do not ask the reader to switch measures. What it delivers is evidence about the dictionary: the tone measure that dominates the statement-communication literature loses resolution precisely on the episodes the paradigm-shift narrative cares about, and any dictionary-based claim about those episodes should be read alongside this measurement-failure evidence.